Reviewer evxp

Due to space limitations, we have uploaded the revision materials to the following anonymous repository:
https://anonymous.4open.science/r/INR_TSD-FC7B/README.md
The newly added figures and tables are also included there.

1. We thank the reviewer for this important suggestion. In our additive setting X(t)=T(t)+S(t) (with an additional residual term), the seasonal component is determined as the residual after trend estimation, i.e., S(t)=X(t)−T(t). In addition, some of the methods we compare against (e.g., moving average, convolution-based filters) only provide trend filtering, with the seasonal component likewise obtained as the residual. For this reason, we emphasized trend recovery in the main text, since accurate trend estimation is the key determinant of overall decomposition quality in this setting.

Please note that we provided the error of seasonality recovery in Table 1 and 3 of the original manuscript for synthetic data set. We will discuss more details about the seasonality recovery in the revised manuscript. Furthermore, we have now conducted further consistency analyses of the recovered seasonality for real SCN data (Table A), demonstrating that our method robustly captures the seasonal component in real data. We will add this in the revised manuscript.

2. We thank the reviewer for this helpful suggestion. We have now provided side-by-side visualizations on synthetic datasets, showing the ground-truth and our method for both trend and seasonality components (Figure. A). We will enhance the quality of the figure and include it in the revised manuscript.

3. We thank the reviewer for this insightful comment. We agree that the connection between the problem formulation and our design choices can be made clearer, and we will strengthen this discussion in the revision.

Our design is guided by three structural aspects of the decomposition problem: (i) the signal is defined in continuous time and can be irregularly sampled, (ii) the seasonal component is inherently oscillatory with unknown frequency, and (iii) the additive decomposition $X(t) = T(t) + S(t)$ introduces potential confounding between components when learned jointly.

First, we adopt implicit neural representations (INR) to model both components as continuous functions of time. This avoids reliance on a fixed temporal grid, naturally handles irregular sampling, and eliminates boundary effects that arise in window-based or convolution-based methods.

Second, we use periodic activations to introduce an inductive bias toward oscillatory structures. This enables the model to represent a wide range of frequencies without requiring explicit specification of the period, in contrast to classical decomposition methods that depend on pre-defined seasonal parameters.

Third, we employ a decoupled (sequential) learning strategy. By first learning the trend and then fitting the residual together with our normalization scheme to separate low and high frequency components, we enforce a separation that stabilizes the decomposition. On the other hand, additive decomposition is not identifiable under joint optimization, as low-frequency components of the seasonality and high-frequency components of the trend can be arbitrarily exchanged.

To support this point, we have now compared the performance between decoupled and joint learning strategies (Table B), showing that the decoupled strategy adopted in our approach outperforms the joint learning strategies adopted in previous approaches. We will add this result in the revised manuscript.

4. We thank the reviewer for this important comment. We agree that the term “tuning-free” should be stated more precisely.

In our work, “tuning-free” specifically refers to the absence of dataset-specific decomposition parameters (e.g., window size, smoothing strength, or pre-specified period), which are typically required in classical decomposition methods. In contrast, our method uses a fixed model architecture and optimization setting across datasets.

We note that this does not imply the absence of hyperparameters altogether, but rather that a single configuration can be used robustly across different datasets. This is enabled by our normalization scheme, which facilitates the separation of trend and seasonality components in a data-adaptive manner (see Appendix C for details). We will clarify this point earlier in the revised manuscript.

To further address the sensitivity of our method to network architecture, we have now conducted additional ablation studies (Table B), showing that the performance remains stable across a wide range of settings, supporting the robustness of our approach.

5. Indeed, we expect similar gains across more backbones or datasets because accurate detrending is critical for forecasting. To support this, we have now investigated additional 6 datasets (Table C), showing that our method consistently improves performance across datasets. We will add this in the revised manuscript.

However, evaluating our method across a broader range of forecasting backbones would require systematically replacing the decomposition module at the front end of decomposition-based forecasting models and takes considerable time. Thus, we will discuss this as the future work in the revised manuscript.

Table A. Comparison of seasonality decomposition performance on 100 SCN datasets. The robustness of seasonality estimation under three downsampling strategies (irregular, sparse, missing) is evaluated using MAE metrics. INR-TSD demonstrates overall superior seasonality decomposition performance. We will further strengthen this comparison by incorporating a wider range of methodologies and evaluation metrics.

Model	Irregular ($MAE \pm STD$)	Sparse ($MAE \pm STD$)	Missing ($MAE \pm STD$)
Ours	**1.69e-3 ± 7.07e-4**	**1.54e-3 ± 6.28e-4**	*1.64e-3 ± 7.29e-4*
STL	3.53e-3 ± 1.04e-3	4.40e-3 ± 1.34e-3	2.95e-3 ± 9.25e-4
RSTL	3.48e-3 ± 1.17e-3	*3.22e-3 ± 1.08e-3*	3.19e-3 ± 1.12e-3
MSTL	*3.52e-3 ± 1.11e-3*	4.22e-3 ± 1.30e-3	**1.50e-3 ± 5.04e-4**
Moving Average	4.24e-3 ± 1.40e-3	4.51e-3 ± 1.46e-3	4.43e-3 ± 1.45e-3

Table B. Ablation results on synthetic datasets used in the original manuscript. We compare the performance of decoupled and joint learning strategies by varying the number of trend and seasonal layer nodes to analyze the sensitivity of the proposed architecture. Lower MSE and MAE indicate better performance. The best results are highlighted in bold.

# Trend Nodes	# Seasonal Nodes	Joint		Decoupled	
		Trend	Seasonality	Trend	Seasonality
		(MSE ± Std / MAE ± Std)	(MSE ± Std / MAE ± Std)	(MSE ± Std / MAE ± Std)	(MSE ± Std / MAE ± Std)
20	20	0.10 ± 0.08 / 0.18 ± 0.09	16.11 ± 8.82 / 3.77 ± 1.15	0.06 ± 0.04 / 0.15 ± 0.08	0.08 ± 0.05 / 0.17 ± 0.08
	40	0.09 ± 0.06 / 0.17 ± 0.07	16.11 ± 8.82 / 3.77 ± 1.16	0.06 ± 0.04 / 0.15 ± 0.08	0.07 ± 0.05 / 0.16 ± 0.08
	60	0.09 ± 0.06 / 0.19 ± 0.09	16.10 ± 8.82 / 3.77 ± 1.16	0.06 ± 0.04 / 0.15 ± 0.08	0.07 ± 0.05 / 0.16 ± 0.08
40	20	0.13 ± 0.08 / 0.22 ± 0.06	16.10 ± 8.82 / 3.77 ± 1.16	**0.05 ± 0.04 / 0.13 ± 0.06**	0.06 ± 0.05 / 0.15 ± 0.06
	40	0.12 ± 0.08 / 0.20 ± 0.08	16.10 ± 8.81 / 3.77 ± 1.16	**0.05 ± 0.04 / 0.13 ± 0.06**	**0.05 ± 0.04 / 0.14 ± 0.06**
	60	0.13 ± 0.08 / 0.22 ± 0.08	16.10 ± 8.82 / 3.77 ± 1.16	**0.05 ± 0.04 / 0.13 ± 0.06**	**0.05 ± 0.04 / 0.14 ± 0.06**
60	20	0.16 ± 0.09 / 0.24 ± 0.09	16.10 ± 8.82 / 3.77 ± 1.16	**0.05 ± 0.05 / 0.13 ± 0.07**	0.06 ± 0.05 / 0.15 ± 0.06
	40	0.12 ± 0.08 / 0.21 ± 0.07	16.10 ± 8.82 / 3.77 ± 1.16	**0.05 ± 0.05 / 0.13 ± 0.07**	**0.05 ± 0.05 / 0.15 ± 0.07**
	60	0.12 ± 0.07 / 0.23 ± 0.10	16.09 ± 8.82 / 3.77 ± 1.16	**0.05 ± 0.05 / 0.13 ± 0.07**	**0.05 ± 0.05 / 0.15 ± 0.07**

Table C. Long-term forecasting results on time series benchmark datasets (ECL, Traffic, Weather, Ettm1, Ettm2, Etth1, and Etth2). We use prediction length O ∈ {96}. A lower MSE indicates better performance. Hereafter, for the tables, the best results are marked in bold and the second optimal in underlined, respectively with MSE/MAE.

Model	ECL	Weather	Ettm1	Ettm2	Etth1	Etth2
	MSE/MAE	MSE/MAE	MSE/MAE	MSE/MAE	MSE/MAE	MSE/MAE
TEMPO (w/ ours)	**0.176/0.274**	**0.196/0.252**	**0.412/0.414**	**0.179/0.265**	**0.393/0.406**	**0.298/0.351**
TEMPO	0.178/0.276	0.211/0.254	0.438/0.424	0.185/0.267	0.400/**0.406**	0.301/0.353
GPT2	0.193/0.288	0.226/0.274	0.486/0.438	0.193/0.273	0.400/0.416	0.320/0.363
T5	0.185/0.282	0.217/0.271	0.529/0.464	0.190/0.268	0.400/0.409	0.328/0.366
PatchTST	0.489/0.546	0.247/0.301	0.733/0.554	0.273/0.345	0.57/0.518	0.379/0.412
Timesnet	0.293/0.369	0.247/0.295	0.518/0.470	0.202/0.290	0.407/0.423	0.315/0.362
FEDformer	0.300/0.399	0.292/0.346	0.698/0.553	0.665/0.634	0.509/0.502	0.385/0.426
ETSformer	0.707/0.638	0.453/0.416	1.117/0.678	0.353/0.404	0.469/0.457	0.405/0.428
Informer	0.512/0.531	0.837/0.711	0.880/0.657	0.263/0.360	0.642/0.562	0.704/0.651
DLinear	0.195/0.292	0.212/0.275	0.624/0.522	0.264/0.352	0.414/0.421	0.334/0.389

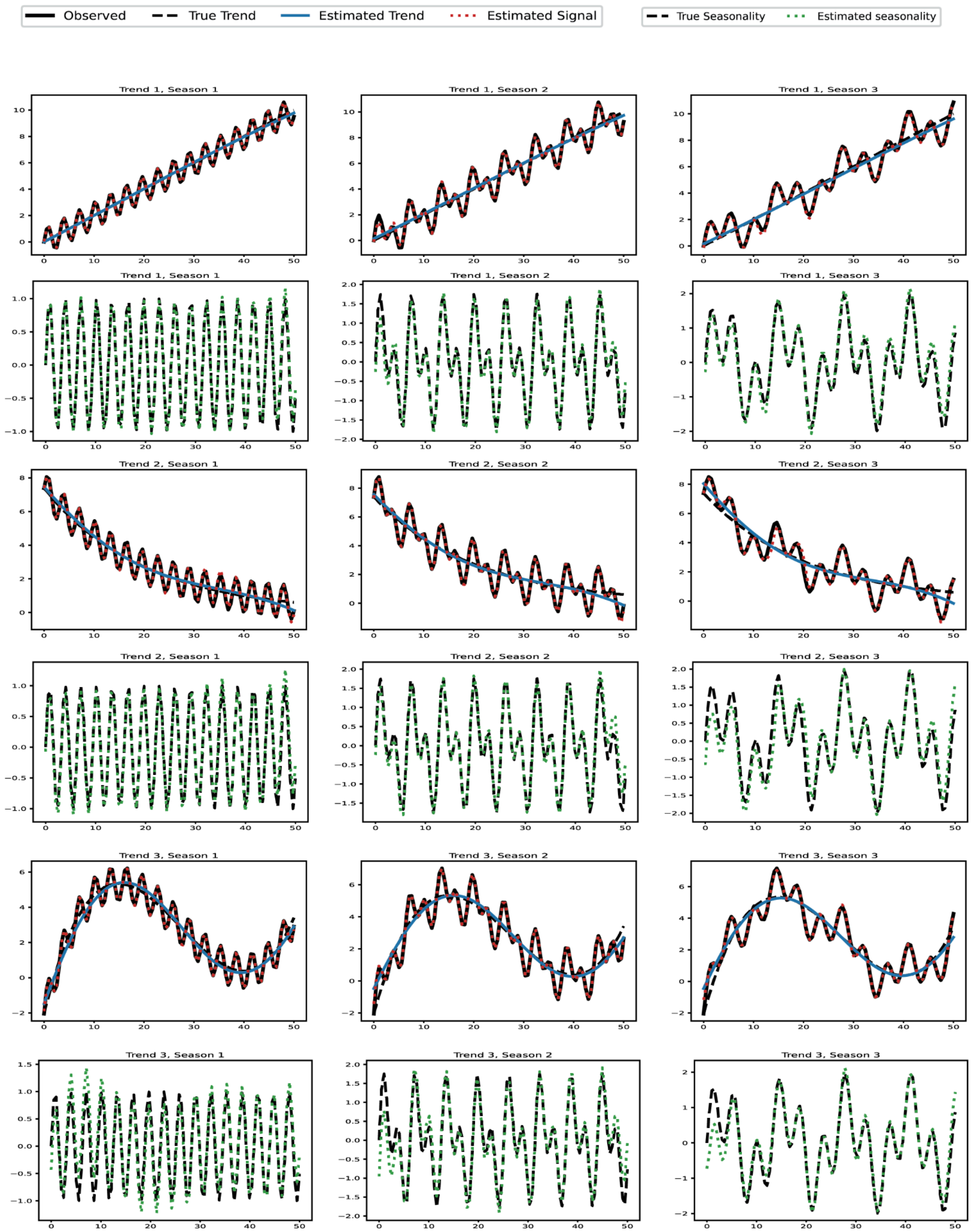

Figure A. Visualizations of the recovered trend and seasonality. Figures (i)–(ix) show side-by-side results on synthetic datasets used in the original manuscript. In each subfigure, the top panel displays the estimated trend along with the sum of the estimated trend and seasonality, which reconstructs the original signal. The bottom panel illustrates the seasonality estimation performance. Overall, INR-TSD effectively captures both trend and seasonality.

Reviewer T9LZ

Due to space limitations, we have uploaded the revision materials to the following anonymous repository:
https://anonymous.4open.science/r/INR_TSD-FC7B/README.md
The newly added figures and tables are also included there.

1. We thank the reviewer for this important question.

First, we clarify that INR-TSD differs from architectures such as N-BEATS and SCNN in both objective and design. N-BEATS and SCNN are primarily forecasting architectures, where structured components such as trend or seasonality are incorporated as internal mechanisms to improve prediction. In particular, while the interpretable version of N-BEATS provides a form of trend-seasonality decomposition, it does not aim to develop a decomposition methodology itself. Rather, decomposition is introduced to make forecast outputs interpretable, using predefined basis expansions (e.g., polynomial and Fourier bases). Similarly, SCNN provides a structured decomposition into multiple components, but this decomposition is embedded within a forecasting framework and optimized for predictive performance.

In contrast, INR-TSD is designed to address time-series decomposition itself as the primary task and evaluates its accuracy and robustness independently of forecasting.

At the modeling level, our approach differs in three key aspects:

- **Sequential decomposition**: trend is estimated first, followed by seasonality on the detrended signal, reducing interference between trend and seasonality components
- **Implicit representation as a continuous function**: both components are modeled as continuous functions rather than discrete basis expansions
- **Component-specific normalization**: distinct inductive biases are introduced through a component-specific normalization**,** where the trend and seasonal components are normalized separately. This facilitates the separation of low- and high-frequency dynamics within a unified INR framework

Second, regarding optimization without ground-truth trend/seasonality: our method does not require supervision of individual components. The separation emerges from the sequential decomposition procedure together with the distinct inductive biases introduced by component-specific normalization. Under our normalization scheme, the trend component is first biased toward capturing smooth, slowly varying (low-frequency) behavior. Subsequently, by applying a component-specific normalization to the detrended residual, the seasonal model is guided to capture higher-frequency oscillatory components (see also Section 3.1. in the original manuscript).

Finally, we agree that the loss terms were not clearly defined in the original manuscript. In the revision, we will explicitly introduce the loss functions, e.g.,

$$L_t = \frac{1}{N}\sum_{j=1}^{N} ||\widehat{T}(t_j) - X_j||, \qquad L_s = \frac{1}{N}\sum_{j=1}^{N} ||\widehat{S}(t_j) - (X_j - \widehat{T}(t_j))||,$$

and clarify their roles in the revised manuscript.

2. We thank the reviewer for this question and agree that the experimental setup was not sufficiently described.

Synthetic data generation

The synthetic datasets are constructed as (see Section 2.1)

$$X(t) = T(t) + S(t) + R(t)$$

where $T(t)$ is a slowly varying trend and $S(t)$ represents seasonality with the residual $R(t)$. For each dataset, we sample 100 time points uniformly over $[0, 50]$ and define three trend functions (linear, exponential, and polynomial, e.g. $T_2(t) = exp(-0.05t + 2)$) and three seasonal functions with different frequency compositions (e.g. $S_3(t) = \sin(0.5\,t) + \sin(\sqrt{2}t)$). The synthetic signals are generated by combining these components (see Figure 2). We will add explicit formulas for T and S in Figure 2.

Downsampling procedure

We consider three types of degraded sampling:

(i) Sparse sampling: we uniformly subsample 40% of the points from the original regularly sampled time series, resulting in evenly spaced but lower-resolution observations.

(ii) Irregular sampling: we randomly select 70% of the original time indices and then sort them, producing non-uniform temporal spacing.

(iii) Missing intervals: we remove a contiguous segment corresponding to 7% of the total observations. In our setting with 100 time points, this corresponds to removing 7 consecutive points (e.g., from the 41st to the 47th indices), simulating structured data loss.

Real irregular data

We agree that evaluation on real irregularly sampled datasets would further strengthen the results. While our current goal is to assess robustness under controlled settings with known ground truth, we have now performed an additional test on a real-world irregularly sampled dataset (Figure B).

In this evaluation, we have treated the trend and seasonality estimated from the full dataset as a reference and have assessed consistency by re-estimating the trend and seasonality using only the first half of the data and comparing the two results. In addition, we have compared our method with pyBOAT (frequency-based) and RobustSTL (a recently proposed decomposition method), and have shown that INR-TSD outperforms both baselines.

The results indicate that INR-TSD reliably recovers both trend and seasonality in irregularly sampled real-world data. We will include these results in the revised manuscript and further expand the comparison to additional baseline methods.

SCN dataset

For the SCN dataset, we clarify that ground-truth components are not available. Therefore, we evaluated consistency and biological plausibility, rather than quantitative decomposition error. We will revise the manuscript to more clearly describe this evaluation setting and distinguish it from synthetic experiments where ground truth is available.

3. We have now extended our forecasting experiments to include six additional benchmark datasets (Table C). Across these datasets, we observe consistent performance improvements when integrating INR-TSD into the same forecasting backbone, indicating that the gains are not limited to a specific subset of data. We will include these additional results in the revised manuscript to strengthen the empirical evidence.

We also agree that reporting standard deviations is important for assessing statistical significance. In the revised version, we will include standard deviations over multiple runs for all forecasting results.

Regarding the scope of evaluation, we clarify that our goal is not to propose a new forecasting architecture, but to develop a general-purpose decomposition module that can be integrated into existing pipelines. In this context, our experiments are designed to isolate and evaluate the effect of improved trend–seasonality decomposition within a fixed forecasting setting. A comprehensive comparison across all recent forecasting architectures—including those with complex time-frequency mechanisms—would require systematically modifying each model to incorporate our decomposition module at the front end, which is beyond the scope of this work. Instead, we position our method as complementary to such approaches rather than a direct alternative. We will revise the manuscript to clarify this scope and improve the positioning of our work relative to recent forecasting models.

4. We thank the reviewer for this helpful suggestion and agree that recent trend/frequency-aware methods should be better discussed in the revised manuscript. In particular, we will expand the related-work section to include recent spectral approaches such as TSLANet and WPMixer in forecasting.

From a decomposition perspective, Fourier- and wavelet-based approaches are well-established and have been extensively studied in prior work. In particular, RobustSTL (Wen et al., IJCAI 2019) explicitly compares trend filtering performance with wavelet-based approaches and demonstrates strong performance as a decomposition method. Based on this, we include RobustSTL as a representative and competitive baseline for frequency-aware decomposition in our experiments. In addition, we include pyBOAT, which employs frequency-based filtering (e.g., low-pass filtering), further covering frequency-based approaches. Therefore, while we do not include all variants of spectral methods, we believe that the included baselines reasonably represent this class of approaches.

Furthermore, many spectral approaches are primarily designed for regularly sampled time series and may require interpolation or preprocessing when applied to irregular or missing observations. In contrast, our method is based on continuous implicit neural representations, which naturally operate in continuous time and can directly handle irregular, sparse, and missing observations without requiring resampling. This enables our approach to extend frequency-aware decomposition to more general data settings where observation patterns are uneven or incomplete. We will revise the related work section to better clarify this positioning and to expand the discussion of related spectral methods.

Romero, David W., et al. "Ckconv: Continuous kernel convolution for sequential data." *arXiv preprint arXiv:2102.02611* (2021).

J. Deng et al., "Disentangling Structured Components: Towards Adaptive, Interpretable and Scalable Time Series Forecasting," in IEEE Transactions on Knowledge and Data Engineering, vol. 36, no. 8, pp. 3783-3800, Aug. 2024, doi: 10.1109/TKDE.2024.3371931.

Oreshkin, Boris N., et al. "N-BEATS: Neural basis expansion analysis for interpretable time series forecasting." *arXiv preprint arXiv:1905.10437* (2019).

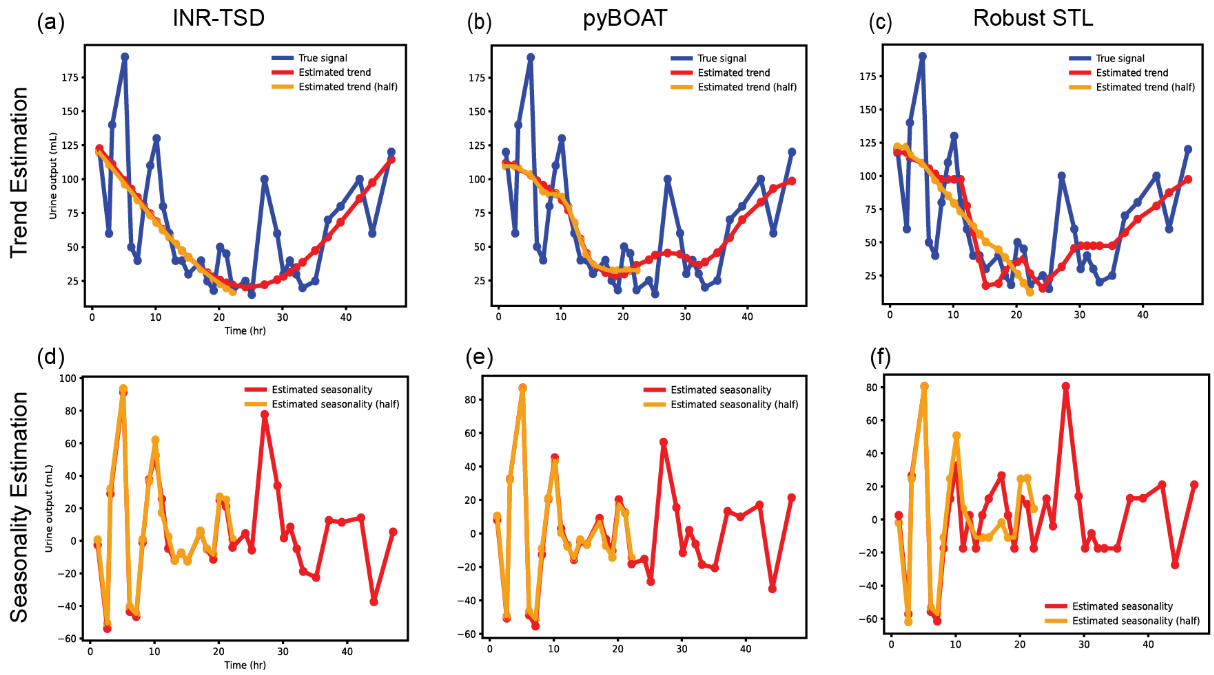

Figure B. Performance of INR-TSD on irregularly sampled real data. INR-TSD, together with the baseline methods pyBOAT and RobustSTL, is evaluated on a real-world dataset consisting of urine measurements collected from patients for the ICU mortality prediction task (https://physionet.org/content/challenge-2012/1.0.0/). This dataset is irregularly sampled and does not provide ground-truth trend or seasonality components. We assess trend ((a)–(c)) and seasonality ((d)–(f)) estimation by comparing results obtained from the full time series (red) and from the first half of the time series (orange). INR-TSD achieves a stable qualitative separation of trend and seasonal components. Note that the seasonality shown in (e) for pyBOAT corresponds to a detrended signal, as the method does not explicitly provide a seasonality decomposition.

Table C. Long-term forecasting results on time series benchmark datasets (ECL, Traffic, Weather, Ettm1, Ettm2, Etth1, and Etth2). We use prediction length O ∈ {96}. A lower MSE indicates better performance. Hereafter, for the tables, the best results are marked in bold and the second optimal in underlined, respectively with MSE/MAE.

Model	ECL	Weather	Ettm1	Ettm2	Etth1	Etth2
	MSE/MAE	MSE/MAE	MSE/MAE	MSE/MAE	MSE/MAE	MSE/MAE
TEMPO (w/ ours)	**0.176/0.274**	**0.196/0.252**	**0.412/0.414**	**0.179/0.265**	**0.393/0.406**	**0.298/0.351**
TEMPO	0.178/0.276	0.211/0.254	0.438/0.424	0.185/0.267	0.400/**0.406**	0.301/0.353
GPT2	0.193/0.288	0.226/0.274	0.486/0.438	0.193/0.273	0.400/0.416	0.320/0.363
T5	0.185/0.282	0.217/0.271	0.529/0.464	0.190/0.268	0.400/0.409	0.328/0.366
PatchTST	0.489/0.546	0.247/0.301	0.733/0.554	0.273/0.345	0.57/0.518	0.379/0.412
Timesnet	0.293/0.369	0.247/0.295	0.518/0.470	0.202/0.290	0.407/0.423	0.315/0.362
FEDformer	0.300/0.399	0.292/0.346	0.698/0.553	0.665/0.634	0.509/0.502	0.385/0.426
ETSformer	0.707/0.638	0.453/0.416	1.117/0.678	0.353/0.404	0.469/0.457	0.405/0.428
Informer	0.512/0.531	0.837/0.711	0.880/0.657	0.263/0.360	0.642/0.562	0.704/0.651
DLinear	0.195/0.292	0.212/0.275	0.624/0.522	0.264/0.352	0.414/0.421	0.334/0.389

Reviewer sFuH

Due to space limitations, we have uploaded the revision materials to the following anonymous repository:
https://anonymous.4open.science/r/INR_TSD-FC7B/README.md
The newly added figures and tables are also included there.

1. We thank the reviewer for this insightful question. In our framework, the effect of ω_0 is handled through the normalization scheme (see also Section 3.1 and Appendix B). Specifically, after normalization, the effective frequency domain is constrained to [0,1], and the frequency weights of the model are initialized uniformly within this range. This allows the model to capture relevant frequency components within a bounded domain, thereby reducing sensitivity to the choice of ω_0. Empirically, we observe that the decomposition performance is stable with a fixed setting ω_0 across datasets (see Figure 3 and Table 1), supporting the use of a default configuration without dataset-specific tuning.

To further investigate this, we have now performed an ablation study analyzing the impact of ω_0 on decomposition performance (Table D), demonstrating that our default setting ($\omega_0 = 1$) is optimal. We will include this result in the revised manuscript.

2. We thank the reviewer for this comment. We have now added two additional experiments: one examining various periodic spectra (Figure C) and the other investigating different sampling rates (Figure D). The results demonstrate that INR-TSD consistently performs well across a range of periodic spectra and remains stable across different sampling rates. This can be attributed to the normalization scheme, which constrains the frequency range of the time series to [0, 1], and to the ability of implicit neural representations to model arbitrary real-valued frequencies rather than being limited to integer-frequency basis functions (e.g., sin(nx)). Furthermore, as continuous functions, implicit neural representations are inherently robust under downsampled data conditions. We will add this in the revised manuscript.

3. We thank the reviewer for this important question. We agree that the current plug-and-play validation is limited in scope. To strengthen this point, we include additional forecasting results on another dataset, where replacing the decomposition module with INR-TSD again yields consistent improvement. This suggests that the observed benefit is not specific to a single dataset.

Regarding backbones, we selected TEMPO because it includes an explicit STL-based decomposition stage, making it a particularly suitable testbed for evaluating whether INR-TSD can serve as a drop-in replacement with minimal architectural change. In contrast, many forecasting architectures do not expose a clearly separated decomposition module, so a direct plug-and-play substitution is not always well-defined. For this reason, our current experiment is intended as a proof of compatibility with decomposition-based forecasting pipelines, rather than a comprehensive benchmark across all backbones. We will revise the manuscript to clarify this scope more explicitly and discuss the comparison with more backbones as a future work.

4. We thank the reviewer for this important question. In general, an additive decomposition of the form $X(t)=T(t)+S(t)$ is not identifiable without additional assumptions. However, under a commonly adopted assumption that the trend and seasonality correspond to slowly varying and oscillatory components, respectively—that is, under a time-scale (or equivalently, frequency) separation assumption where the trend occupies low-frequency components and the seasonality captures higher-frequency oscillations—the decomposition becomes effectively identifiable.

Our method is explicitly designed to enforce this separation through a structured decomposition procedure. In particular, we adopt a decoupled learning strategy in which the trend is first extracted as the non-oscillatory (low-frequency) component, and the seasonality is then modeled from the detrended residual. In addition, the normalization scheme (Appendix C) regularizes both the scale and frequency content of each component.

Together, these mechanisms introduce inductive biases that discourage overlapping solutions and promote a clear allocation of low- and high-frequency behaviors. This leads to a stable and practically identifiable decomposition under the time-scale separation assumption.

We will revise the manuscript to clarify this point and explicitly state the role of these assumptions and design choices. Providing formal identifiability guarantees beyond such separation assumptions remains an important direction for future work.

Table D. Impact of ω_0 on decomposition performance. The impact SIREN's ω_0 is investigated by evaluating the performance of INR-TSD under different weight initialization intervals $[0, \omega_0]$. The results of this ablation study show that our default setting outperforms configurations where ω_0 deviates from 1. This indicates that our normalization scheme effectively constrains the frequency values within the range $[0, 1]$.

ω_0	0.5	1 (Default)	2	4	10	20	30
$MAE \pm STD$	0.246 $\pm$ 0.387	**0.131** $\pm$ **0.051**	0.139 $\pm$ 0.055	0.143 $\pm$ 0.035	0.140 $\pm$ 0.042	0.334 $\pm$ 0.244	0.366 $\pm$ 0.230

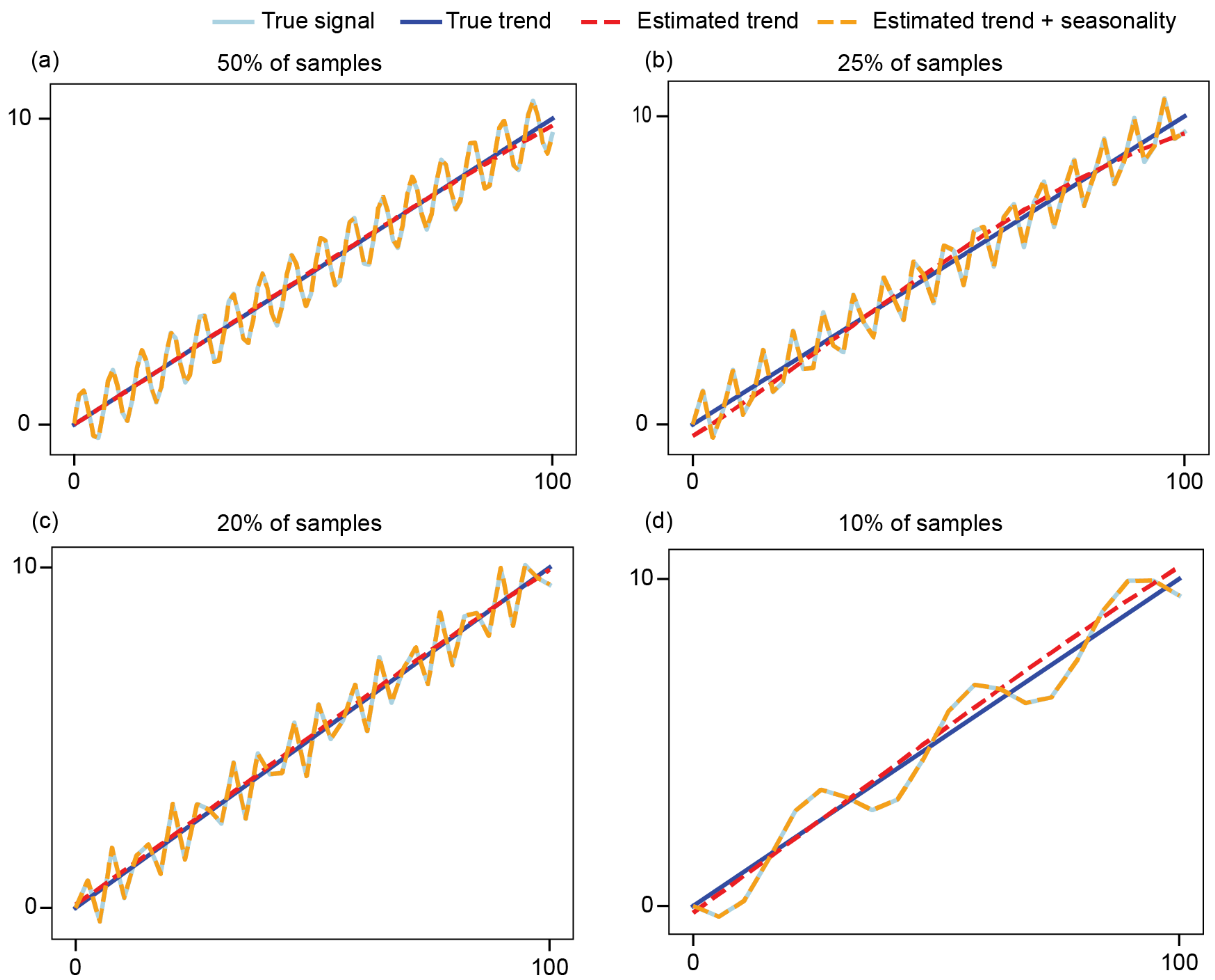

Figure D. Performance of INR-TSD on synthetic data with various sampling rates. For the second experiment, we fixed a predefined combination of trend 0.1x and seasonality sin(x) as shown in Fig. C-(b), and progressively reduced the sampling rate (originally 200 points over the time interval (0, 100)) from (a) 50% to (b) 25%, (c) 20%, and (d) 10%. Across all sampling rate settings, INR-TSD effectively recovers the true trend (blue) with its estimate (red), and reconstructs the overall signal (light blue) as the sum of the estimated trend and seasonal components (orange). These results indicate that the performance of INR-TSD remains stable across a range of sampling rates.

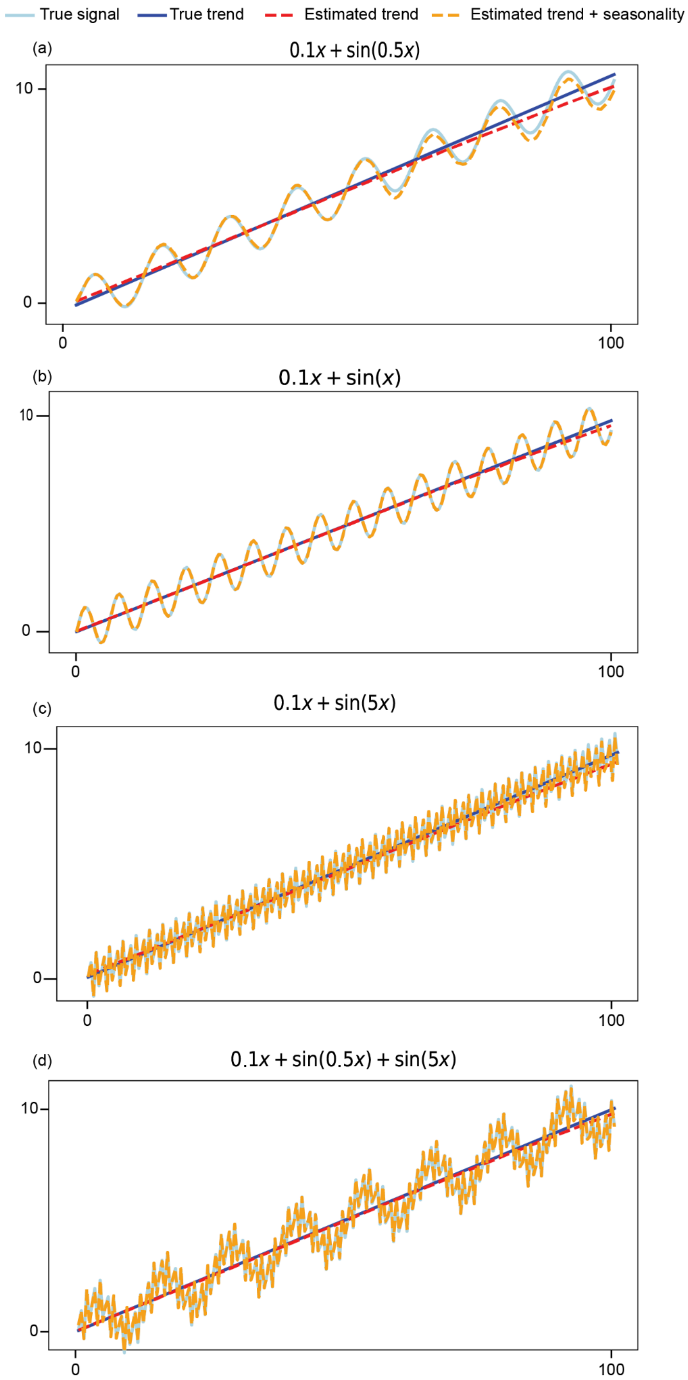

Figure C. Performance of INR-TSD on synthetic data with various periodic spectra. The trend component is fixed as 0.1x, while the seasonal component is constructed using different frequencies. Specifically, the seasonality components are defined as (a) sin(0.5x), (b) sin(x), (c) sin(5x), and (d) sin(0.5x) + sin(5x). Across all frequency settings, INR-TSD effectively recovers the true trend (blue) with its estimate (red), and reconstructs the overall signal (light blue) as the sum of the estimated trend and seasonal components (orange). These results demonstrate that INR-TSD performs consistently well across a wide range of periodic spectra.

Reviewer sbRk

Due to space limitations, we have uploaded the revision materials to the following anonymous repository:
https://anonymous.4open.science/r/INR_TSD-FC7B/README.md
The newly added figures and tables are also included there.

1. We would like to clarify that our primary focus is on time series decomposition, rather than developing a new state-of-the-art forecasting model. Accordingly, the baselines in our experiments are selected to demonstrate (i) decomposition quality and (ii) plug-and-play compatibility with representative forecasting models.

For forecasting, we include a recent and widely used model (TEMPO) to illustrate that our method can be integrated into existing pipelines as a plug-and-play decomposition module. Our goal is not to exhaustively benchmark against all recent forecasting architectures, but to demonstrate that the proposed decomposition module is broadly applicable.

While we agree that benchmarking against a broader set of recent forecasting models could be informative, such an extensive comparison is beyond the primary scope of this work, as it would shift the focus from decomposition to forecasting model benchmarking. Our goal is instead to establish INR-TSD as a general-purpose decomposition module that can be readily applied across models. That said, we will clarify this scope more explicitly in the revised manuscript and discuss extending the evaluation to additional state-of-the-art models as future work. Instead, we have now evaluated our plug-and-play performance across a broader range of datasets (Table C).

2. We would like to clarify that the forecasting dataset is explicitly specified as ETTh2 in Section 4.5. While this information is already included in the current version, we acknowledge that it may not have been sufficiently prominent, and we will revise the manuscript to make it more clearly identifiable.

To further address the reviewer's concern about dataset diversity, we have now conducted additional forecasting experiments on six additional datasets, where we observed consistent improvements when replacing the decomposition module with our method (Table C). These results suggest that the observed benefit is not limited to a single dataset. We will add this in the revised manuscript.

Taken together, we would like to emphasize that our primary focus is on time series decomposition, and the forecasting experiment is intended to demonstrate plug-and-play compatibility rather than to provide a comprehensive benchmarking across all forecasting models and datasets. We will revise the manuscript to better clarify this scope.

3. We would like to first clarify that the evaluation procedure for trend estimation was described in the paper (Section 4.3 and Figure 4), where the trend recovery process and its evaluation under downsampling are illustrated.

Specifically, the trend estimation accuracy is evaluated on synthetic data by comparing the recovered trend with the ground truth under different downsampling settings. This evaluation protocol follows prior work on time series decomposition (Q. Wen et al., AAAI, 2019). We acknowledge that this procedure may not have been sufficiently explicit or easy to follow in the current presentation.

To address this, we will revise the manuscript to more clearly describe the evaluation protocol and improve the visibility of these details, including additional methodological clarification and references where appropriate.

4. We would like to clarify that our work focuses on time series decomposition, and the paper already includes comparisons with representative decomposition methods to evaluate decomposition quality and robustness in both synthetic and real data (see e.g. Figure 2, 3, 5 and Table 1).

Regarding forecasting performance, this concern is closely related to the previous point. As clarified above, our forecasting experiments are intended to demonstrate plug-and-play compatibility rather than to provide an exhaustive comparison across all forecasting models. In addition, the forecasting baselines used in our experiments are directly adopted from those originally included in the Tempo framework, ensuring a fair and consistent comparison setting.

We agree that the positioning of our method relative to related approaches can be further clarified, and we will revise the manuscript to make this more explicit.

5. We agree that the current presentation may give the impression of broader applicability than what is directly validated in the experiments.

Our primary contribution is a time series decomposition framework that is robust to downsampling (e.g., irregularly sampled data, missing intervals, and sparse observations) and does not require dataset-specific tuning. The mentions of anomaly detection and representation learning are included only as potential future directions of the proposed decomposition framework, rather than as claims that these tasks are addressed or validated in the current work (see Appendix A).

In the current paper, we focus on decomposition quality and demonstrate its practical utility through forecasting as a representative downstream task. We will revise the manuscript to more clearly scope these statements and avoid overgeneralization.

Table C. Long-term forecasting results on time series benchmark datasets (ECL, Traffic, Weather, Ettm1, Ettm2, Etth1, and Etth2). We use prediction length O ∈ {96}. A lower MSE indicates better performance. Hereafter, for the tables, the best results are marked in bold and the second optimal in underlined, respectively with MSE/MAE.

Model	ECL	Weather	Ettm1	Ettm2	Etth1	Etth2
	MSE/MAE	MSE/MAE	MSE/MAE	MSE/MAE	MSE/MAE	MSE/MAE
TEMPO (w/ ours)	**0.176/0.274**	**0.196/0.252**	**0.412/0.414**	**0.179/0.265**	**0.393/0.406**	**0.298/0.351**
TEMPO	<u>0.178</u>/<u>0.276</u>	<u>0.211</u>/<u>0.254</u>	<u>0.438</u>/<u>0.424</u>	<u>0.185</u>/<u>0.267</u>	<u>0.400</u>/**0.406**	<u>0.301</u>/<u>0.353</u>
GPT2	0.193/0.288	0.226/0.274	0.486/0.438	0.193/0.273	<u>0.400</u>/0.416	0.320/0.363
T5	0.185/0.282	0.217/0.271	0.529/0.464	0.190/0.268	<u>0.400</u>/<u>0.409</u>	0.328/0.366
PatchTST	0.489/0.546	0.247/0.301	0.733/0.554	0.273/0.345	0.57/0.518	0.379/0.412
Timesnet	0.293/0.369	0.247/0.295	0.518/0.470	0.202/0.290	0.407/0.423	0.315/0.362
FEDformer	0.300/0.399	0.292/0.346	0.698/0.553	0.665/0.634	0.509/0.502	0.385/0.426
ETSformer	0.707/0.638	0.453/0.416	1.117/0.678	0.353/0.404	0.469/0.457	0.405/0.428
Informer	0.512/0.531	0.837/0.711	0.880/0.657	0.263/0.360	0.642/0.562	0.704/0.651
DLinear	0.195/0.292	0.212/0.275	0.624/0.522	0.264/0.352	0.414/0.421	0.334/0.389